✅ Product Requirements Document

Product: Kedu — A Strict, Proactive Financial Agent on WhatsApp

Owner: Safaya Ikechukwu **Status:** Draft v4 **Type:** Personal tool (v1) — not a startup, not for external users **Stack:** n8n (orchestration) + Rubies Microfinance Bank (account/ledger) + WhatsApp Cloud API (interface) + AI model (classification/reasoning/chat) + Cursor (custom logic)

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1. Problem Statement

I make good money, but I cannot account for where it goes. Spending happens in small, frequent, impulsive bursts — ₦5k here, ₦1k there — with no friction between the impulse and the transaction. By the time I check my balance, I cannot reconstruct the pattern of how the money left.

This isn't a visibility problem. Bank apps and statements already show me *what* happened after the fact. The real problem is that **there is nothing between the impulse and the transfer** — no pause, no second opinion, no rule enforcing my own stated intentions.

The stakes are concrete: I want to close this year having moved toward a landed property. That requires sustained, disciplined saving — which has never been a failure of my income, only of my spending behavior.

Root causes

1. **No friction at the point of spend.** Transfers execute instantly with zero checkpoint.
2. **No pattern memory.** Nothing tracks *why* money leaves, only that it did.
3. **No enforced budgeting.** Budgets exist as intentions, not as system-level constraints.
4. **No proactive accountability.** I only reflect on spending when I go looking for it.

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2. Solution Overview

Kedu is a financial agent that lives on WhatsApp, backed by a real bank account (Rubies

Microfinance Bank), that sits **between me and my money** — not just reporting on it.

Everyday spending money moves into the Kedu account, and the agent becomes the only path through which outbound transfers happen. This makes rules enforceable rather than advisory, because Kedu isn't observing a transaction after the fact — it's the thing authorizing it.

Core philosophy: daily unlocks with a fixed spending window

Kedu enforces spending discipline through **daily allowance unlocking, spendable only within a fixed daily window — 11AM to 8PM.** Only today's allowance is available, and it's only freely accessible during that window. The rest of the week's money remains locked until its scheduled day arrives — the same way a weekly paycheck doesn't hand you Friday's money on Monday.

Controlled overrides (the flexibility layer, deliberately costly):

- **Early access** — spending *outside* the 11AM–8PM window (before 11am or after 8pm) — is possible, but incurs a fee of **₦1,000 + 0.2%** of the amount accessed. This is the override for wanting to move money at the "wrong" hour, not for wanting to spend more than planned.
- **Borrowing against the week's allowance:** possible, at **₦1,000 + 0.4%** of the amount — *unless* there are unused funds left over from the previous day, in which case that surplus is drawn against first, free of charge.
- **Weekly rollover, within the week only:** unused daily allowance rolls forward as free surplus within the same week — but resets at the week boundary. Leftover funds do not carry over into the following week.
- **Settings changes:** early access, borrowing, and exit all share the same 7-day minimum before any change request is even eligible for review.

This replaces any idea of a smart "emergency override" — verifying whether a recipient is a hospital versus a restaurant in real time isn't reliably solvable (there's no public API that classifies Nigerian bank accounts by vendor category), and building a judgment-based bypass door defeats its purpose. The override path stays fee-based and deliberately annoying rather than judgment-based — cost is the friction, not a gatekeeper trying to guess intent.

What Kedu also does

Beyond gatekeeping transfers, Kedu builds a financial profile at onboarding, helps make sense of money already spent (via statement upload or open banking), and proactively talks to the user about their money like an advisor would — not just a locked box.

Income-tier branching (critical logic, not just data capture)

Financial profiling at onboarding must actually change what Kedu recommends, not just collect numbers for display. Two branches matter:

- **Spending problem vs. income problem.** If someone's income genuinely doesn't leave meaningful room to restrict further (e.g. a low income in a high-cost city), locking their spending down harder isn't discipline — it's just imposing hardship. Kedu needs to recognize this case and shift its posture from "restrict spending" to "grow income" — surfacing that distinction honestly to the user rather than defaulting every profile into a restriction plan.
- **Flat salary vs. irregular income.** A percentage-based budget (spend no more than X% of whatever comes in) fits founders, freelancers, and business owners with irregular cash flow far better than a flat daily/weekly naira figure, which assumes a steady monthly salary. The income-type question already captured at onboarding (salaried/founder/multiple streams) should actually drive which budgeting math gets used, not just sit as profile metadata.

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3. Full Feature Set

#	Feature	Description
1	WhatsApp-native interface	100% of interaction happens on WhatsApp — no separate app
2	Rubies-powered bank account	Real account issuance via Rubies Microfinance Bank
3	Daily unlocks & weekly rollover, within a fixed window	Only today's allowance is available, spendable freely during the 11AM–8PM window. The rest remains locked until scheduled. Unused daily allowance rolls forward within the week as free surplus, but resets at the week boundary — no carryover into the next week.
4	Controlled overrides	Early access (spending before 11am or after 8pm): ₦1,000 + 0.2% fee. Borrowing against the week's allowance: ₦1,000 + 0.4% fee, waived if unused funds from the previous day cover it. Changes to these settings can only be made after 7 days.
4a	Incoming funds notification	When Kedu receives funds into the account, it sends a WhatsApp message confirming receipt

5	Pre-transfer confirmation	Before any transfer, agent confirms who the money is going to and shows today's remaining allowance
6	Recipient tagging (manual, since no reliable classification API exists)	New recipients get tagged by the user on first send (rent / food / vendor / other) and remembered going forward
7	Financial profiling at onboarding	Captures income, income type (salaried/founder/multiple streams), assets (incl. crypto/stablecoins), and recommends a budget split method based on proven frameworks (e.g. 50/30/20) matched to the user's actual situation
7a	Income-tier branching	If income doesn't leave meaningful room to restrict, Kedu shifts posture toward income-growth guidance rather than defaulting to a restriction plan. Irregular/business income uses percentage-based budgeting instead of a flat daily/weekly figure.
8	Statement upload & declutter	Upload account statements or financial documents and Kedu parses and organizes them into readable spending summaries
9	Open banking connection	Alternative to manual upload — connect other bank accounts directly for a full picture
10	Conversational rule-setting	Chat with Kedu in natural language to define spending rules and budgets
11	Spend analysis & recommendations	Kedu identifies overspend categories and suggests concrete alternatives
12	Large-transfer calls	Kedu calls the user (not just messages) when a large amount is about to be sent, as an extra checkpoint
13	Proactive financial updates	Unprompted nudges and updates on financial standing
14	Tax assistance	Helps make sense of tax obligations from transaction history
15	Savings wallets / sub-buckets	Money ring-fenced into named buckets (rent, blocked savings, etc.) with independent release rules
16	Monthly proactive breakdown	Unprompted monthly summary of spending and savings, sent via WhatsApp/call
17	Voice-first onboarding	Onboarding happens entirely on WhatsApp, voice-note-first, with a friendly-but-probing conversation (see Section 7)

18	Settings-change escalation (“the managers”)	Users cannot change core settings (e.g. the time-lock) directly. They must email/call a “manager” — still AI, but a distinct, more formal persona — who references their original stated goal and progress, and makes them explain why they want to change course before anything is altered
19	Calendar integration (deferred)	Proactively cross-references upcoming events against remaining budget — good future idea, explicitly not part of MVP

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4. v0 Scope — Must-Haves Only

This is the MVP — matches the original “Kedu v1” feature list exactly: AI onboarding, dedicated Kedu account, daily unlocks with weekly rollover, controlled overrides, and deliberate exit.

4.1 Must-have

- Rubies account provisioned and connected to WhatsApp number — every user gets a dedicated account; salary and transfers land there
- AI onboarding: understand why the user joined, create a personalized spending plan, let the user choose a strictness level, explain the commitment before continuing
- **Daily unlocks & weekly rollover, within a fixed window:** only today’s allowance is available, spendable freely during 11AM–8PM; the rest stays locked until scheduled; unused daily allowance rolls forward as free surplus within the week
- **Controlled overrides:** early access (spending before 11am or after 8pm) costs ₦1,000 + 0.2%; borrowing against the week’s allowance costs ₦1,000 + 0.4%, waived if unused funds from the previous day cover it; changes to these settings can only be made after 7 days; unused funds rest at the end of the week and cannot be carried over into the next week
- **Deliberate exit:** disabling restrictions requires a cooling-off period and human (Manager persona) review before the account returns to normal, after seven days
- **Pre-transfer confirmation:** every transfer gets a WhatsApp confirmation (recipient, amount, tag, remaining allowance) before it executes
- **Manual recipient tagging:** new recipients get tagged by the user on first send and stored for future transfers

- **Incoming funds notification:** when Kedu receives funds into the account, it sends a WhatsApp message confirming receipt
- **Basic financial profiling at onboarding:** income, income type, and a recommended budget split — feeds the daily/weekly numbers
- **Income-tier branch:** at minimum, detect when income doesn't leave meaningful restriction room and shift to income-growth messaging instead of forcing a restriction plan; use percentage-based budgeting (not flat figures) for irregular/business income
- Simple persistent storage for recipient tags, transaction log, allowance/rollover state, and override history (Postgres or Supabase via n8n)

4.2 v0.5 — Commercial version (after 10 beta testers)

Once v0 is proven on real beta users, this tier adds:

- Verify-transaction button
- Email signup and onboarding email sequences
- Expense breakdown (daily, weekly, monthly views)
- Fee consequences shown clearly (what an override actually cost, over time)
- Multiple-choice questions rendered as tappable buttons instead of typed replies, where WhatsApp supports it
- Airtime purchases + cashback

4.3 Why this scope

The single behavior to test first: **does daily-unlock-within-a-window, backed by a genuinely annoying override fee, actually stop impulsive spending — or does the override just become the new normal?** Everything else — statement intelligence, chat-based rules, recommendations, calls — only matters once this core discipline mechanism is proven to hold under real pressure.

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5. Technical Notes

- **Orchestration:** n8n handles webhook-in (WhatsApp message) → window check (11AM–8PM) → allowance check → override-fee logic if applicable → recipient tagging/confirmation → Rubies API call → WhatsApp response-out.
- **Window + allowance enforcement:** simplest possible implementation — a scheduled

daily unlock plus a time-of-day check in n8n, no AI needed for this part. Keep the core discipline mechanism dumb and unbreakable rather than clever.

- **Rollover & override tracking:** daily spend logged against today's allowance; unused amounts roll forward as surplus within the week; override fees (₦1,000 + 0.2% or ₦1,000 + 0.4%) calculated and confirmed before any early-access or borrowed transfer executes; settings-change requests timestamped against a 7-day minimum before they're eligible for review.
- **Statement parsing:** PDF/CSV upload → AI model extracts and categorizes transactions → structured summary returned via WhatsApp.
- **Money flow constraint:** Kedu can only govern transfers that originate *from* the Rubies account. Actual spending money needs to be routed into that account for the daily-unlock model to mean anything.
- **Data store:** lightweight — recipient tags, transaction log, budget state, parsed statement data, timestamps. Supabase or Postgres via n8n is sufficient for v1.
- **Trust/verification moment:** at account verification (BVN or equivalent), Kedu must explain in plain language why this information is needed and what it isn't used for — written as part of the character/tone work, not generic legal copy. This is the highest-risk point for a non-technical user to feel surveilled or bail out.

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6. Onboarding Flow

See the companion document **Kedu-Onboarding-Script.md** for the full stage-by-stage conversation script (Discovery → Welcome → Identity → Money Story voice note → AI Reflection → Structured Questions → Plan → Account Creation → Funding → First Transaction → Habit-Forming Follow-up).

Key structural principles carried through the script:

- One question at a time, never front-loaded
- Push back gently on vague answers rather than accepting deflection ("I don't know" gets a follow-up, not a move-on)
- Voice notes are the primary input for the deeper reflection questions; structured multiple-choice/button questions are used later for profiling data (strictness preference, spending categories) where buttons are actually faster than typing
- If a voice note is unclear, Kedu plays back what it understood and asks for confirmation rather than guessing

- The account number and funding ask come only *after* real reflection and a proposed plan — not as the opening transaction
- Kedu means "how are you" in Igbo — this is used as a small, real piece of identity in the welcome message, not just trivia

Terminology used consistently: the problem Kedu addresses is described as "impulse spending and overeating" — this pairing is used every time, not shortened to spending alone.

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7. Settings & the "Manager" Escalation Mechanic

Core rule: **users cannot change core settings (like the daily allowance, rollover rules, or override fees) directly.** Any change request routes to "the managers" — a distinct AI persona, more formal and procedural in tone than Kedu itself, reachable by email or call. **Changes can only be made after 7 days** — this minimum holds regardless of how the request is framed.

Why this works

The friction isn't a button that says "are you sure?" — it's the act of having to write down or say out loud *why* you want to break your own rule, to something that isn't going to just say yes. That's a meaningfully higher bar than an in-app confirmation dialog.

How it should behave

- **Tone contrast is deliberate.** The manager persona should feel more procedural and less warm than Kedu — even though it's the same underlying AI system — because that contrast is what makes it feel like an actual internal process rather than Kedu talking to itself.
- **References specifics, not generic pushback.** The callback should quote the user's own stated goal and progress from onboarding — date joined, amount saved, what they said they wanted.
- **A real delay is part of the design.** Not an instant reply — even a few hours of "we'll get back to you" gives the impulse room to pass on its own before the setting actually changes.
- **The verification step stays soft-touch.** A cooling-off conversation about the reason, not a hard classification system.
- **Dependency on onboarding data.** This mechanic only works if the original stated goal

was actually captured during onboarding — the emotional weight comes from quoting the user's own words back to them.

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8. Success Criteria for v1

This is working if, after 4–6 weeks of real use:

- The daily-unlock-within-a-window model has actually prevented spending that would otherwise have happened impulsively — and the override fee is enough friction that it isn't being paid routinely just to work around the system.
- Spending can be reconstructed for a given week without manually digging through bank apps.
- The system has not been disabled, bypassed, or abandoned out of frustration.

If those three hold, it's worth investing in v0.5 (verify-transaction button, email sequences, expense breakdowns, fee-consequence visibility, button-based questions, airtime + cashback) and beyond. If not, the daily-unlock model itself needs rethinking before layering more on top of it.